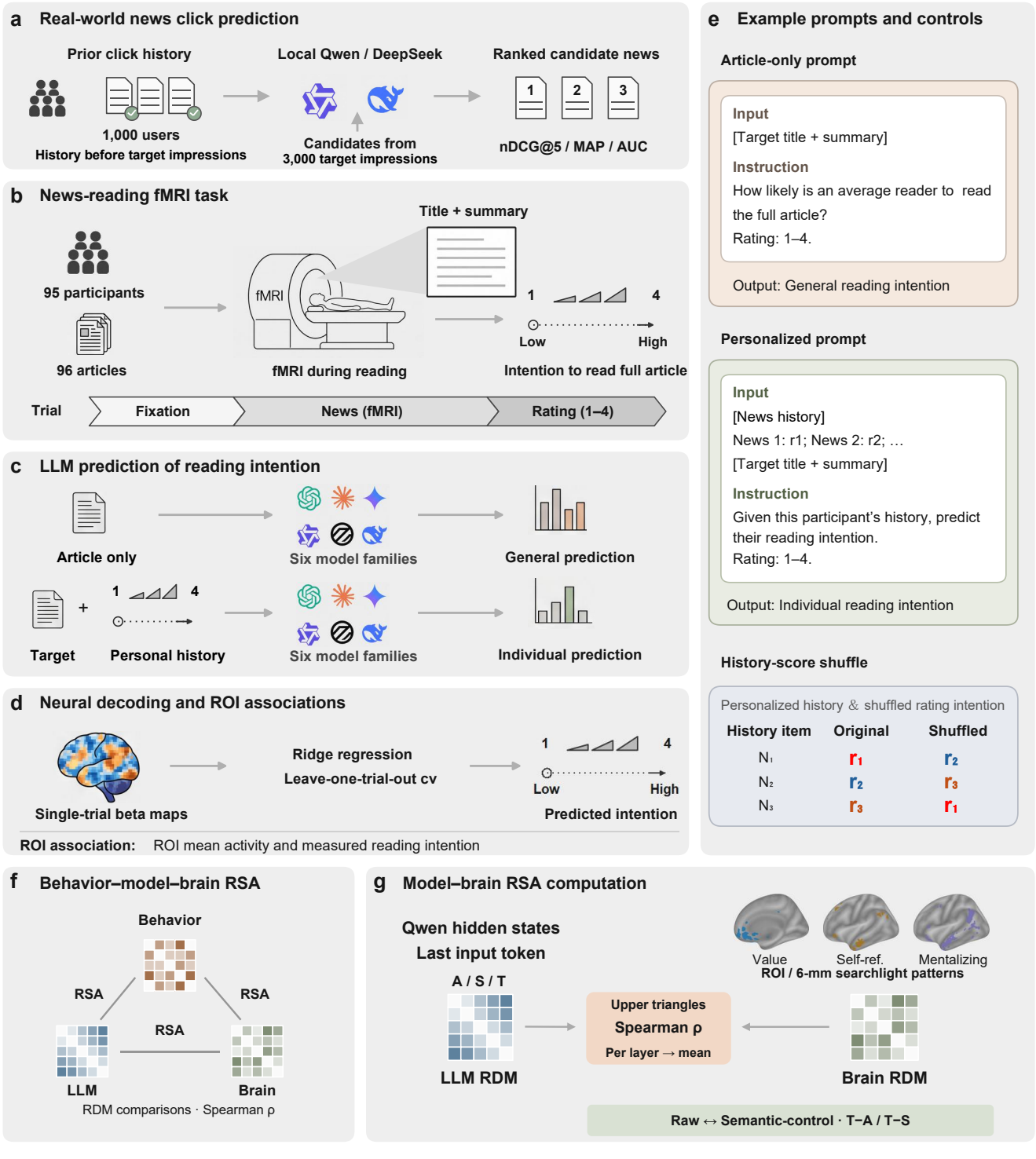

a
Real-world news click prediction
Prior click history
1,000 users
History before target impressions
Local Qwen / DeepSeek
Candidates from
3,000 target impressions
Ranked candidate news
1
2
3
nDCG@5 / MAP / AUC
b
News-reading fMRI task
95 participants
96 articles
Title + summary
fMRI
fMRI during reading
1
4
Low
High
Intention to read full article
Trial
Fixation
News (fMRI)
Rating (1–4)
c
LLM prediction of reading intention
Article only
Six model families
General prediction
Target
Personal history
Six model families
Individual prediction
d
Neural decoding and ROI associations
Single-trial beta maps
Ridge regression
Leave-one-trial-out cv
Low
High
Predicted intention
ROI association:
ROI mean activity and measured reading intention
e
Example prompts and controls
Article-only prompt
Input
[Target title + summary]
Instruction
How likely is an average reader to read the full article?
Rating: 1–4.
Output: General reading intention
Personalized prompt
Input
[News history]
News 1: r1; News 2: r2; …
[Target title + summary]
Instruction
Given this participant's history, predict their reading intention.
Rating: 1–4.
Output: Individual reading intention
History-score shuffle
Personalized history & shuffled rating intention
History item
Original
Shuffled
N1
r1
r2
N2
r2
r3
N3
r3
r1
f
Behavior–model–brain RSA
Behavior
RSA
RSA
RSA
LLM
Brain
RDM comparisons · Spearman ρ
g
Model–brain RSA computation
Qwen hidden states
Last input token
A / S / T
LLM RDM
Upper triangles
Spearman ρ
Per layer → mean
Value
Self-ref.
Mentalizing
ROI / 6-mm searchlight patterns
Brain RDM
Raw ↔ Semantic-control · T−A / T−S